Addressing the global misinformation crisis one flag at a time

RedFlag

The future of fact checking

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Old Dominion University
CS 410
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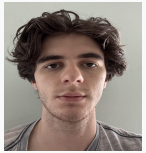
Our *team*



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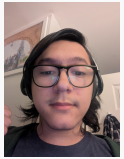

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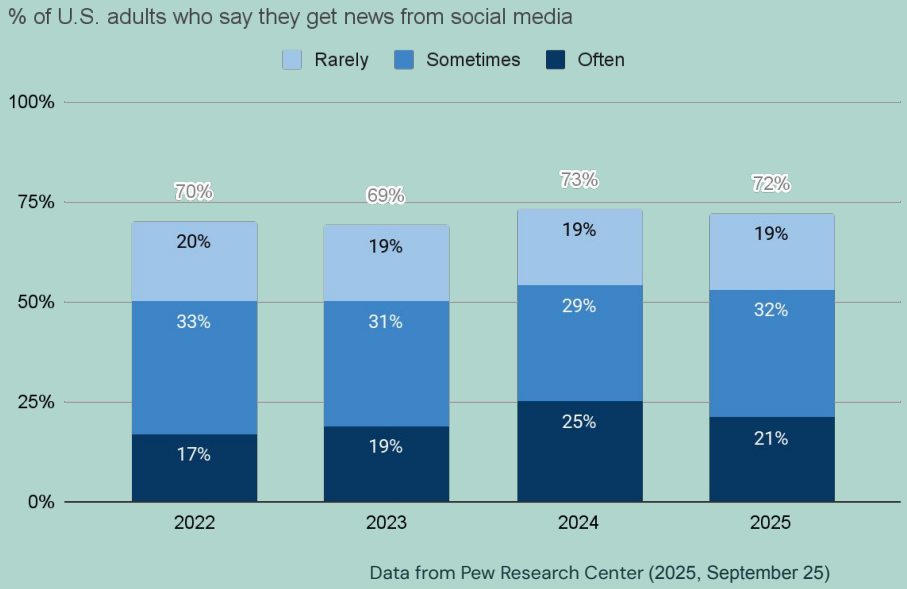
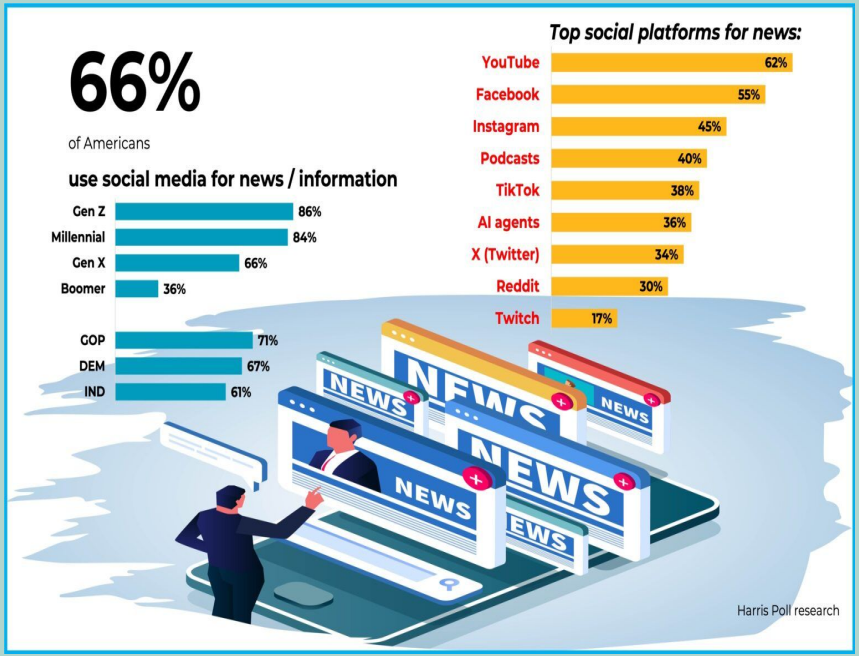
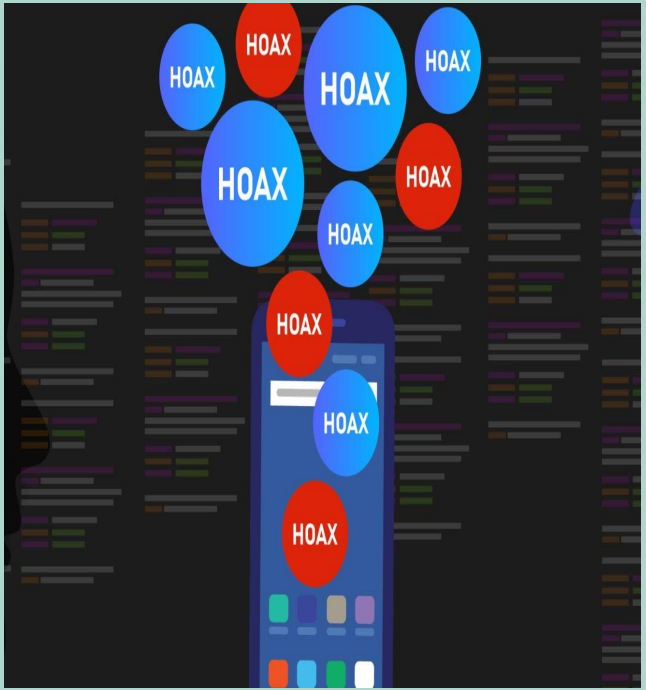
Social Media as News



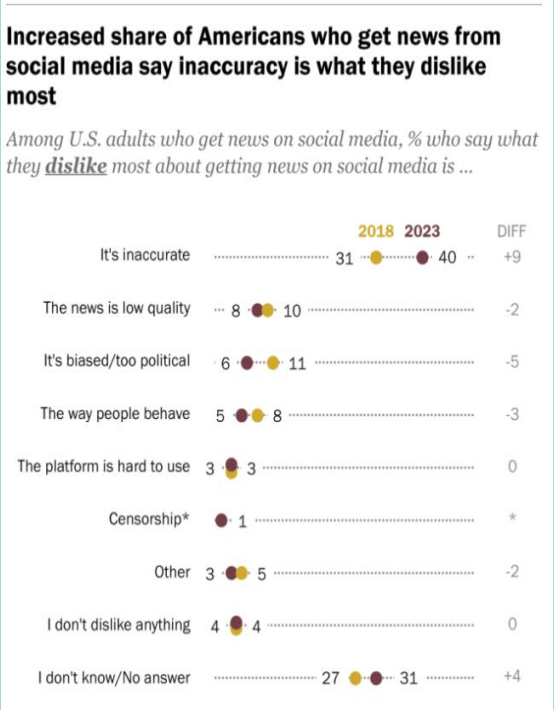


Misinformation in our Digital Ecosystem



- 94% of people report encountering misinformation online (The Alan Turing Institute, 2024)
- 38.2% of U.S. news consumers report unknowingly sharing fake news. (Redline Digital, 2024)

Distrust is growing



Pew Research Center. (2024, February 7)

- Concerns about inaccurate information on social media are rising (Pew Research Center, 2024)
- 40% of Americans who get news from social media cite inaccuracy as their biggest concern. (Pew Research Center, 2024)
- Only 37% say they trust information from social media platforms. (Pew Research Center, 2023)

So why does it persist?

- People don’t pay attention to what they’re consuming/sharing (Ordway, D 2021)
- Social media platforms reward engagement over accuracy (Ceylan, G 2023)

Evidence shows it is primarily a mindless action, done out of passivity and without care

- Community based fact-checking is too slow, corrections don't come in until long after posts have taken off (Blanding, M 2025)
- Professional human fact-checking is also costly
- However, purely AI based fact checking often only compounds the problem as AI checkers have been found to be wrong 37-94% of the time (Ford 2025)
- Going to a separate website is time consuming and requires extra effort

Current Fact-checkers aren't effective

Why does this matter?

When misinformation spreads:

- Public trust declines (Pew Research Center, 2023)
- Users feel concerned but lack effective tools (Pew Research Center, 2024)
- False information spreads unintentionally (Redline Digital, 2024)

The fact checking tools of today are either too niche or bothersome to tackle this misinformation epidemic.

RedFlag

Problem Characteristics (THE GAP)

Inconvenience and Disconnected Verification

Fact-checking often requires manual searching outside the platform.

Lack of Credibility Awareness

Users often receive no real-time credibility signals while browsing.

Motivation to Verify

Platforms and algorithms encourage instant sharing without friction current fact checking tools require a break from media consumption to use.

Lack of Universal Tool

Many existing verification tools are platform-specific and not unified.

Bias Algorithms

Users algorithms feed them content they inherently agree with which makes misinformation easier to spread.

The Solution

RedFlag is a cross-platform browser extension that automatically detects and flags potentially misleading content in real time, providing credibility indicators and source links directly within the user's browsing experience.

What it Will Do

How will RedFlag work?

Integrated Detection

- Integrated by a chrome extension to scan content automatically

Real-Time Detection

- Highlights questionable information immediately

Frictionless & Complete

- Highlights require no user input
- Can deliver detailed overview on request

RedFlag

Closing THE GAP

Convenience	Awareness	Motivation	Fragmentation
Real Time Integrated Detection	Real-Time detection	Frictionless Detection System	Multi-Platform capabilities

What it Will Not Do

What will RedFlag's Limitations be?

Unbiased political alignment

- RedFlag will not support any specific political standpoint
- No ideological labeling
- Evidence based evaluation

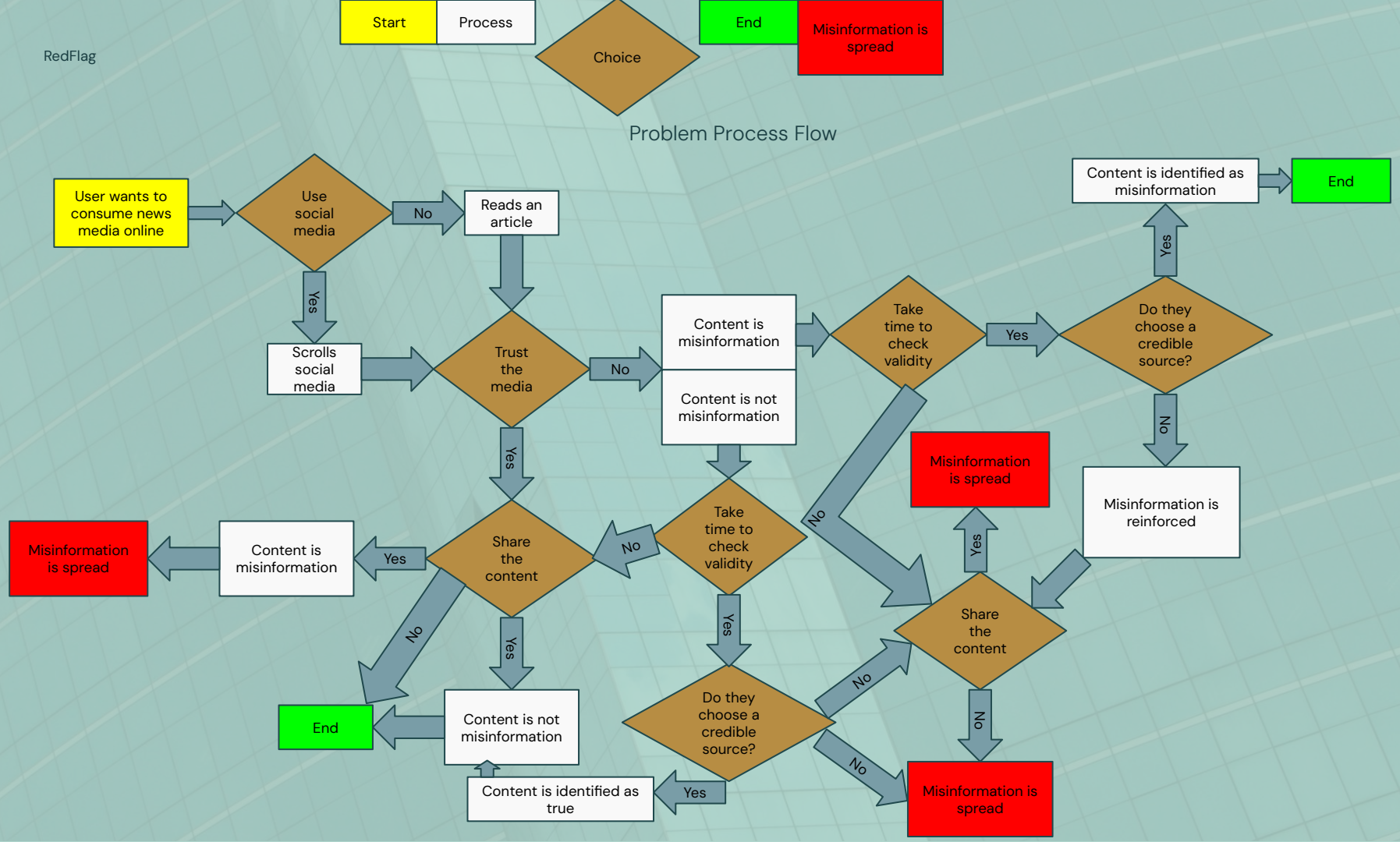
Uninterrupted flow

- Pop-ups would interrupt user's browsing experience
- Highlighting text is significantly less intrusive

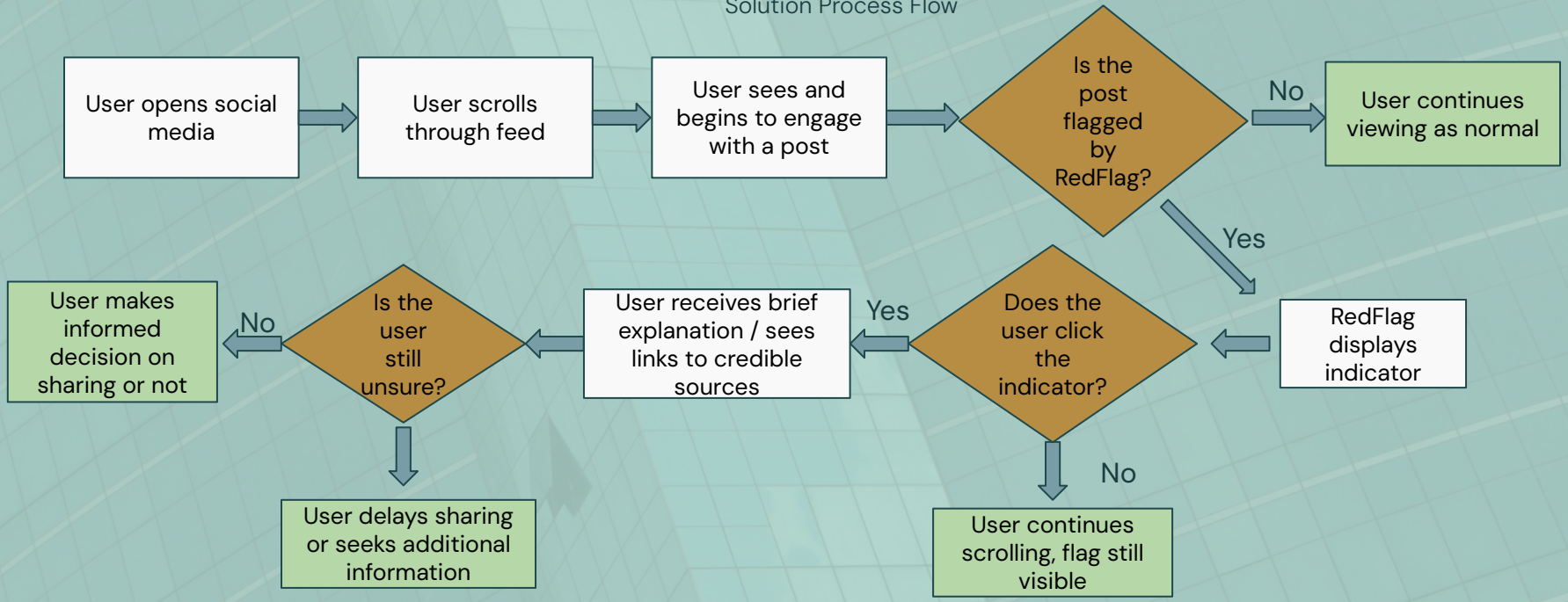
Does NOT replace professional fact-checkers

- Supports decision-making without dictating it
- Optional context is provided upon the user's request
- Supports journalism, but does not substitute it

RedFlag


Start
Process
Choice
End
Misinformation is spread
Problem Process Flow
User wants to consume news media online
Use social media
No
Reads an article
Yes
Scrolls social media
Trust the media
No
Content is misinformation
Content is not misinformation
Take time to check validity
Yes
Do they choose a credible source?
Yes
Content is identified as misinformation
End
No
Misinformation is reinforced
Yes
Misinformation is spread
No
Share the content
No
Misinformation is spread
Yes
Take time to check validity
No
Share the content
Yes
Do they choose a credible source?
No
Misinformation is spread
Yes
Content is identified as true
Content is misinformation
Yes
Misinformation is spread
Share the content
No
End
Yes
Content is not misinformation

Solution Process Flow



Competitors vs Redflag

	Fact checker databases (Example: Google Fact Check Explorer)	News Guard	Community Notes	AI
FEATURES	• Search for stories by keyword, person, or image that have been debunked • Fact check other stories yourself	• Rates trustworthiness of a news source on a scale from 1-100 • Team of journalists fact checking	• Allows users to add contextual explanations or fact checks to posts • Other contributors rate notes for helpfulness, only helpful notes are shown	• Ask AI the veracity of a claim/source (e.g. @Grok is this true?) • AI uses its dataset to respond
STRENGTHS	• Quickly able to search for rumors or claims being made online • Aggregate source of online disinformation	• Real human input on trustworthiness • Partnered with organizations for better fact checking	• Posts remain visible with extra context added • Uses a diverse group of users to reduce bias	• Ease of use- users already familiar with service, easily available, immediate response • Low manpower required
WEAKNESSES	• Allows biased sources to fact check other sources • Does not provide input on every claim made online • No verified source of claims	• Possible bias due to human fact checkers	• Slow response time makes notes appear long after a misleading post has spread • Organized groups can attempt to manipulate note ratings	• Frequently incorrect • Confidently wrong, fooling users • Dataset is fallible
	VS Fact checker databases	VS News Guard	VS Community Notes	VS AI
WHAT REDFLAG DOES BETTER / DIFFERENT	• Rather than search for stories, it will be real time when reading articles	• Works more actively to inform the user rather than passively on standby to check	• With the browser extension it will work for more than just online posts, and be faster due to a centralized database	• Will use AI solely increasing work flow of fact checking, but not act as a source of authority on any subject matter

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Development Tools

Languages

Test Frameworks

Version Control

Documentation

Major Functional Components

Browser Extension UI

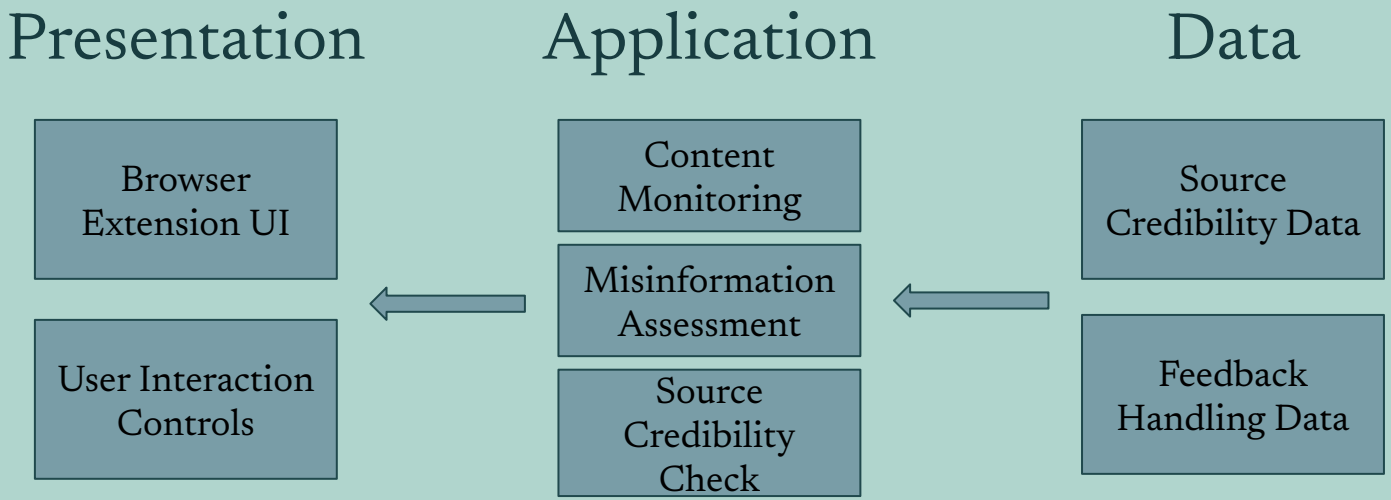
User Interaction Controls

Content Monitoring

Misinformation Assessment

Source Credibility Data

Feedback Handling Data



Major Functional Components Diagram

Glossary

- Friction - something that causes an application to not be seamless while working.

Risks

Appendix

Real World Product vs Prototype Table

Elevator Pitch

RedFlag is a browser extension designed to reduce the spread of online misinformation. It runs in the background while users browse social media and news sites and automatically highlights potentially false or misleading content directly on the screen. Instead of relying on users to fact-check information themselves, RedFlag brings awareness to misinformation at the moment it's encountered. With one click, users can view a short explanation and links to credible sources, helping them make informed decisions before believing or sharing content.

The problem plaguing online news

Despite the vastly popular concern in regarding digital inaccuracy, social media users frequently propagate misinformation because existing verification methods are inconvenient and require active user intervention. There is a critical need for a passive, real-time tool that bridges the gap between content consumption and fact-checking for daily social media users.

Who does this affect?

- Users of social media
  - Teens
    - 85% of young people now get their news primarily from social media. (Stagwell, 2025)
  - 55+
    - On Twitter, 80% of fake news was shared by users over age 50 (Marshall, 2025)